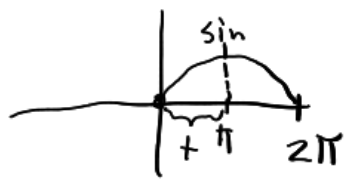


3. v polarnem zapisu:



a) $z = 1 - \cos \alpha + i \sin \alpha ; \alpha \in (0, 2\pi)$

$$|z| = \sqrt{(1 - \cos \alpha)^2 + \sin^2 \alpha}$$

$$= \sqrt{1 - 2\cos \alpha + \cos^2 \alpha + \sin^2 \alpha}$$

$$= \sqrt{2 - 2\cos \alpha} = \sqrt{\frac{2(1 - \cos \alpha)}{2}}$$

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$= 2 \sin \frac{\alpha}{2}$$

lahko, ker obstaja na intervalu $(-\frac{\pi}{2}, \frac{\pi}{2})$

$$\tan \varphi = \frac{\sin \alpha}{1 - \cos \alpha} = \frac{2 \cos \frac{\alpha}{2} \sin \frac{\alpha}{2}}{2 \sin^2 \frac{\alpha}{2}} = \cot \frac{\alpha}{2} = \tan \left(\frac{\pi}{2} - \frac{\alpha}{2} \right)$$

$$\varphi = \frac{\pi}{2} - \frac{\alpha}{2}$$

$$z = 2 \sin \frac{\alpha}{2} e^{i \left(\frac{\pi}{2} - \frac{\alpha}{2} \right)}$$

b) $z = \frac{1 + i \tan \frac{\varphi}{2}}{1 - i \tan \frac{\varphi}{2}}$

$$|z|^2 = z \bar{z} = \frac{1 + i \tan \frac{\varphi}{2}}{1 - i \tan \frac{\varphi}{2}} \cdot \frac{1 - i \tan \frac{\varphi}{2}}{1 + i \tan \frac{\varphi}{2}} = 1$$

$$z = \frac{(1 + i \tan \frac{\varphi}{2})^2}{1 + \tan^2 \frac{\varphi}{2}} = \frac{1 + 2i \tan \frac{\varphi}{2} - \tan^2 \frac{\varphi}{2}}{1 + \tan^2 \frac{\varphi}{2}}$$

$$= \frac{1}{\cos^2 \frac{\varphi}{2}}$$

$$= \cos^2 \frac{\varphi}{2} - \tan^2 \frac{\varphi}{2} \cos^2 \frac{\varphi}{2} + i 2 \tan \frac{\varphi}{2} \cos^2 \frac{\varphi}{2}$$

$$\operatorname{Re}(z) = \cos^2 \frac{\varphi}{2} - \sin^2 \frac{\varphi}{2} = \cos \varphi$$

$$\operatorname{Im}(z) = 2 \tan \frac{\varphi}{2} \cos^2 \frac{\varphi}{2} = 2 \frac{\sin \frac{\varphi}{2}}{\cos \frac{\varphi}{2}} \cos^2 \frac{\varphi}{2} = 2 \sin \frac{\varphi}{2} \cos \frac{\varphi}{2} = \sin \varphi$$

$$\tan \varphi = \frac{\sin \varphi}{\cos \varphi}$$

$$\varphi = \varphi$$

$$z = e^{i\varphi}$$

4) Reži, skiziraj

a) $z^3 = 1$

$$z = |z| e^{i\varphi}$$

$$z^3 = |z|^3 e^{3i\varphi}$$

$$1 = 1 e^{i2k\pi}$$

$$k \in \mathbb{Z}$$

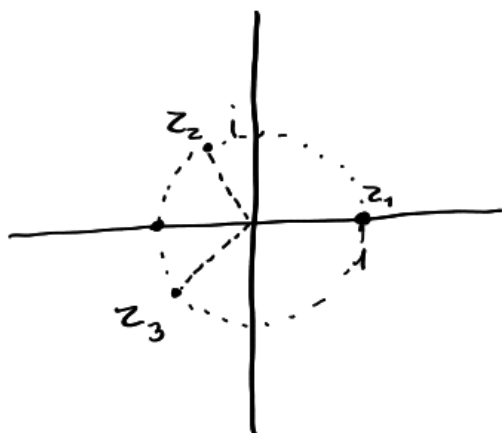
$$|z|^3 = 1$$

$$|z| = 1$$

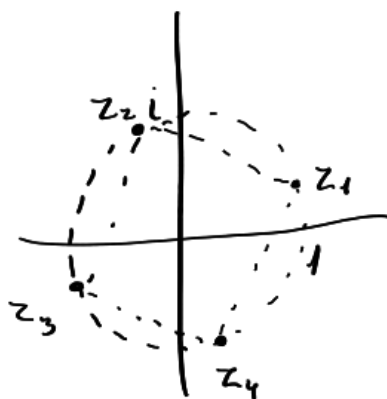
$$3\varphi = 2k\pi$$

$$\varphi = \frac{2}{3} k\pi$$

$$k \in \{0, 1, 2\}$$



b) $z^4 = i$



$$z^4 = |z|^4 e^{4i\varphi}$$

$$i = 1 \cdot e^{i(\frac{\pi}{2} + 2k\pi)}$$

$$|z|^4 = 1$$

$$|z| = 1$$

$$4\varphi = \frac{\pi}{2} + 2k\pi$$

$$\varphi = \frac{\pi}{8} + \frac{k\pi}{2}$$

$$k \in \{0, 1, 2, 3\}$$

$$z_1 = \frac{\pi}{8}$$

$$z_2 = \frac{5\pi}{8}$$

$$z_3 = \frac{9\pi}{8}$$

$$z_4 = \frac{13\pi}{8}$$

7) izpelji ad. izr. za $\cos n\varphi, \sin n\varphi$; $n=3, n=4$

$$z = \cos \varphi + i \sin \varphi$$

$$\boxed{n=3:}$$

$$z^3 = \cos 3\varphi + i \sin 3\varphi$$

$$\begin{aligned} z^3 &= (\cos \varphi + i \sin \varphi)^3 = \cos^3 \varphi + 3 \cos^2 \varphi i \sin \varphi + 3 \cos \varphi i^2 \sin^2 \varphi + \\ &+ i^3 \sin^3 \varphi = \underbrace{\cos^3 \varphi - 3 \cos \varphi \sin^2 \varphi}_{\cos 3\varphi} + i \underbrace{(3 \cos^2 \varphi \sin \varphi - \sin^3 \varphi)}_{\sin 3\varphi} \end{aligned}$$

$$\boxed{n=4:} \quad z = \cos \varphi + i \sin \varphi$$

$$z^4 = \cos 4\varphi + i \sin 4\varphi$$

$$\begin{aligned} z^4 &= (\cos \varphi + i \sin \varphi)^4 = \cos^4 \varphi + 4 \cos^3 \varphi i \sin \varphi + 6 \cos^2 \varphi i^2 \sin^2 \varphi + \\ &+ 4 \cos \varphi i^3 \sin^3 \varphi + i^4 \sin^4 \varphi = \\ &= \underbrace{\cos^4 \varphi - 6 \cos^2 \varphi \sin^2 \varphi + \sin^4 \varphi}_{\cos 4\varphi} + i \underbrace{(4 \cos^3 \varphi \sin \varphi - 4 \cos \varphi \sin^3 \varphi)}_{\sin 4\varphi} \end{aligned}$$

8. $\varphi \in (0, 2\pi)$ $n \in \mathbb{N}$: izračunaj vsoti

$$a) \sum_{k=0}^n \cos k\varphi =$$

$$b) \sum_{k=0}^n \sin k\varphi =$$

$$\sum_{k=0}^n (\cos k\varphi + i \sin k\varphi) = \sum_{k=0}^n e^{ik\varphi}$$

$$S = \frac{a_e (k^{n+1} - 1)}{k - 1} = \frac{1 (e^{i\varphi})^{n+1} - 1}{e^{i\varphi} - 1} = \frac{\cos(n\varphi + \varphi) + i \sin(n\varphi + \varphi) - 1}{\cos \varphi + i \sin \varphi - 1}$$

$$= \frac{(\cos(n\varphi + \varphi) + i \sin(n\varphi + \varphi) - 1)(\cos \varphi - 1 - i \sin \varphi)}{(\cos \varphi - 1)^2 + \sin^2 \varphi} =$$

$$= \frac{\cos(n\varphi + \varphi) \cos \varphi - \cos(n\varphi + \varphi) - i \cos(n\varphi + \varphi) \sin \varphi + i \sin(n\varphi + \varphi) \cos \varphi - i \sin(n\varphi + \varphi) - i^2 \sin(n\varphi + \varphi) \sin \varphi - \cos \varphi + 1 + i \sin \varphi}{\cos^2 \varphi - 2 \cos \varphi + 1 + \sin^2 \varphi}$$

$$= \frac{-i^2 \sin(n\varphi + \varphi) \sin \varphi - \cos \varphi + 1 + i \sin \varphi}{1}$$

$$\operatorname{Re} s = \frac{\cos(n\varphi + \varphi) \cos \varphi - \cos(n\varphi + \varphi) + \sin(n\varphi + \varphi) \sin \varphi - \cos \varphi + 1}{2 - 2 \cos \varphi} =$$

$$= \frac{\cos(n\varphi) - \cos(\varphi(n+1)) - \cos \varphi + 1}{4 \sin^2 \frac{\varphi}{2}}$$

$$= \frac{-2 \sin\left(\frac{n\varphi + \varphi(n+1)}{2}\right) \sin\left(\frac{n\varphi - \varphi(n+1)}{2}\right) + 2 \sin^2 \frac{\varphi}{2}}{4 \sin^2 \frac{\varphi}{2}} = \left[\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2} \right]$$

$$= \frac{\sin \frac{\varphi}{2} + \sin \frac{2n\varphi + \varphi}{2} \sin \frac{\varphi}{2}}{2 \sin^2 \frac{\varphi}{2}} = \frac{1}{2} + \frac{\sin(n\varphi + \frac{\varphi}{2})}{2 \sin \frac{\varphi}{2}} \quad a)$$

$$\operatorname{Im}(s) = \frac{-\cos(n\varphi + \varphi) \sin \varphi + \sin(n\varphi + \varphi) \cos \varphi - \sin(n\varphi + \varphi) + \sin \varphi}{2 - 2 \cos \varphi} =$$

$$= \frac{\sin(n\varphi + \varphi - \varphi) - \sin(n\varphi + \varphi) + \sin \varphi}{4 \sin^2 \frac{\varphi}{2}} = \frac{\sin n\varphi - \sin \varphi(n+1) + \sin \varphi}{4 \sin^2 \frac{\varphi}{2}}$$

5.

a)

$$z^2 - 2iz - 2 = 0$$

$$z_{1,2} = \frac{2i \pm \sqrt{-4 + 4 \cdot 2}}{2} = \frac{2i \pm \sqrt{4}}{2} = \frac{2i \pm 2}{2} = i \pm 1$$

b)

$$z^2 - (1 - 4i)z - 5 - 5i = 0$$

$$z_{1,2} = \frac{1 - 4i \pm \sqrt{(1 - 4i)^2 + 4(5 + 5i)}}{2} = \frac{1 - 4i \pm \sqrt{1 - 8i - 16 - 20 + 20i}}{2} =$$

$$= \frac{1 - 4i \pm \sqrt{5 + 12i}}{2}$$

$$w^2 = 12i + 5$$

$$(x + yi)^2 = 12i + 5$$

$$x^2 + 2xyi - y^2 = 12i + 5$$

$$x^2 - y^2 = 5$$

$$2xy = 12$$

$$x = \frac{6}{y}$$

$$\frac{36}{y^2} - y^2 = 5$$

$$-y^4 - 5y^2 + 36 = 0$$

$$y^4 + 5y^2 - 36 = 0$$

$$(y^2 + 9)(y^2 - 4) = 0$$

$$(y + 2)(y - 2)(y^2 + 9) = 0$$

$$2 \cdot 18$$

$$3 \cdot 12$$

$$4 \cdot 9$$

$$y_1 = 2$$

$$x_1 = 3$$

$$y_2 = -2$$

$$x_2 = -3$$

$$\sqrt{5 + 12i} = \pm (3 + 2i)$$

$$z_{1,2} = \frac{1-4i \pm (3+2i)}{2}$$

$$z_1 = \frac{1-4i+3+2i}{2} = \frac{4-2i}{2} = 2-i$$

$$z_2 = \frac{1-4i-3-2i}{2} = \frac{-2-6i}{2} = -(1+3i)$$

6.

a) $|z|^6 = z^3 + 15z^3 - 64$

$$|z|^6 = |z|^3 e^{-3i\varphi} + 15|z|^3 e^{3i\varphi} - 64$$

$$\varphi = 0:$$

$$|z|^6 = |z|^3 + 15|z|^3 - 64$$

$$|z|^6 - 16|z|^3 + 64 = 0$$

$$(|z|^3 - 8)^2 = 0$$

$$|z|^3 = 8$$

$$|z| = 2$$

$$z_1 = 2$$

$$z_2 = 2e^{i \cdot \frac{\pi}{3}}$$

$$z_3 = 2e^{i \cdot \frac{2\pi}{3}}$$

$$z_4 = 2e^{i\pi}$$

$$z_5 = 2e^{i \cdot \frac{4\pi}{3}}$$

$$z_6 = 2e^{i \cdot \frac{5\pi}{3}}$$

all:

$$z = 2e^{i \frac{k\pi}{3}}$$

$$k \in \{0, 1, 2, 3, 4, 5\}$$

b) $a^n = \bar{a} \cdot n \geq 2$

$$|z|^n e^{in\varphi} = |z| e^{-i\varphi}$$

$$\frac{|z|^n e^{in\varphi}}{|z| e^{-i\varphi}} = 1$$

$$|z|^{n-1} e^{i\varphi(n+1)} = 1$$

$$|z|=1$$

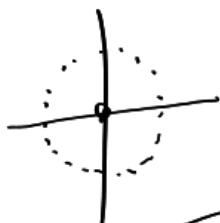
$$e^{i\varphi(n+1)} = 1 = e^{2\pi ki}$$

$$i\varphi(n+1) = 2\pi ki$$

$$\varphi = \frac{2k\pi}{n+1}; k \in \{0, 1, \dots, n\}$$

$$z = 0$$

$$z = e^{i\frac{2k\pi}{n+1}}; k \in \{0, 1, 2, \dots, n\}$$



VEKTORJI V $\mathbb{R}^3$

1.

paralelogram

$$A(-3, -2, 0)$$

$$B(3, -3, 1)$$

$$C(5, 0, 2)$$

$$D(-1, 1, 1)$$

a) dolž. str.

- kot med diagonalama

$$\cos \varphi = \frac{\vec{AC} \cdot \vec{BD}}{|\vec{AC}| \cdot |\vec{BD}|}$$

$$= \frac{-32 + 8}{48} = -\frac{24}{48} = -\frac{1}{2}$$

$$\varphi =$$

$$V = \mathbb{R}^3$$



vekt. prostor

$$F = \mathbb{R}$$



polje (iz polja jemljemo skalarje, s katerimi množimo vektorje)

$$\vec{AB} = (6, -1, 1)$$

$$|\vec{AB}| = \sqrt{38}$$

$$|\vec{BC}| = \sqrt{4+9+1} = \sqrt{14}$$

$$\vec{AC} = (8, 2, 2)$$

$$|\vec{AC}| = \sqrt{72} = 6\sqrt{2}$$

$$\vec{BD} = (-4, 4, 0)$$

$$|\vec{BD}| = 4\sqrt{2}$$

